

Welcome to the Universe

Discover the universe: Learn about the history of the cosmos, what it's made of, and so much more.

EXPLORE:

Exoplanets

Worlds beyond our solar system.

Stars

Giant balls of hot gas that burn for millions to billions of years.

Black Holes

Concentrations of matter with gravity so powerful not even light can escape.

Galaxies

Collections of stars, planets, and vast clouds of gas and dust bound together by gravity.

Swift Boost July 2026

All spacecraft in low Earth orbit experience drag caused by our planet's atmosphere, and after 21 years, the altitude of NASA's Neil Gehrels Swift Observatory has entered a phase of rapid decay due to a bout of increased solar activity. NASA has contracted Katalyst Space to prepare a robotic servicing mission, launching a spacecraft called LINK that will boost Swift to a higher orbit. The project will demonstrate a key capability for the future of space exploration while also extending Swift's scientific life. The boost mission launched on July 3, 2026.

Swift (LINKs Up for a) Boost

On-Orbit Checkouts in Progress for Mission to Boost NASA's Swift

Teams Make Contact With Spacecraft Set to Boost NASA's Swift

Mission To Boost NASA's Swift Launches From Marshall Islands

NASA, Partners Update Launch Date for Mission to Boost Swift

Launch of Swift Boost Mission Delayed, Teams Reviewing Data

Swift Spies on a Snacking Black Hole

When a star strays too close to a monster black hole, gravitational forces can break it apart in a violent event called a tidal disruption. But sometimes the star survives only to come back for more. In 2022, scientists using NASA's Swift spotted a Sun-like star slowly being stripped by a black hole. Every few weeks the star swung close enough to lose material and produce a flare of X-ray light that Swift could see, giving us a rare, repeated glimpse of a black hole that would otherwise be completely invisible.

Swift: The Telescope That Refuses to Quit

When NASA's Neil Gehrels Swift Observatory launched in 2004, it was built for a two-year mission to chase gamma-ray bursts. Twenty-one years later, it has observed not only gamma-ray bursts, but also comets, the first light ever seen from a gravitational wave source, and many other phenomena. When a gyroscope failed in 2024, the team was ready with a backup plan they'd kept in their pocket for 15 years.

NASA's Swift Celebrates 21 Years of Multiwavelength Science

Originally designed as a satellite dedicated to studying GRBs — gamma-ray bursts, the most powerful explosions in the cosmos — Swift also monitors active galaxies powered by supermassive black holes, studies stars undergoing X-ray flares, nova outbursts, and supernova explosions, observes comets and asteroids in our own solar system, and conducts long-term observations of a variety of objects. Swift now occupies a place at the crossroads of multiwavelength, time domain, and multimessenger astronomy.

The Many Scientific Lives of Swift

NASA Missions Probe Game-Changing Cosmic Explosion

NASA's Swift Mission Maps a Star's 'Death Spiral' into a Black Hole

NASA Missions Catch First Light from a Gravitational-Wave Event

NASA's Newly Renamed Swift Mission Spies a Comet Slowdown

NASA's Swift Produces Best Ultraviolet Maps of the Nearest Galaxies

Discover More Topics From NASA

